Logarithmic conformal field theory with boundaries

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joint work with

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Outline

- Bulk and boundary CFT
- Algebraic reformulation

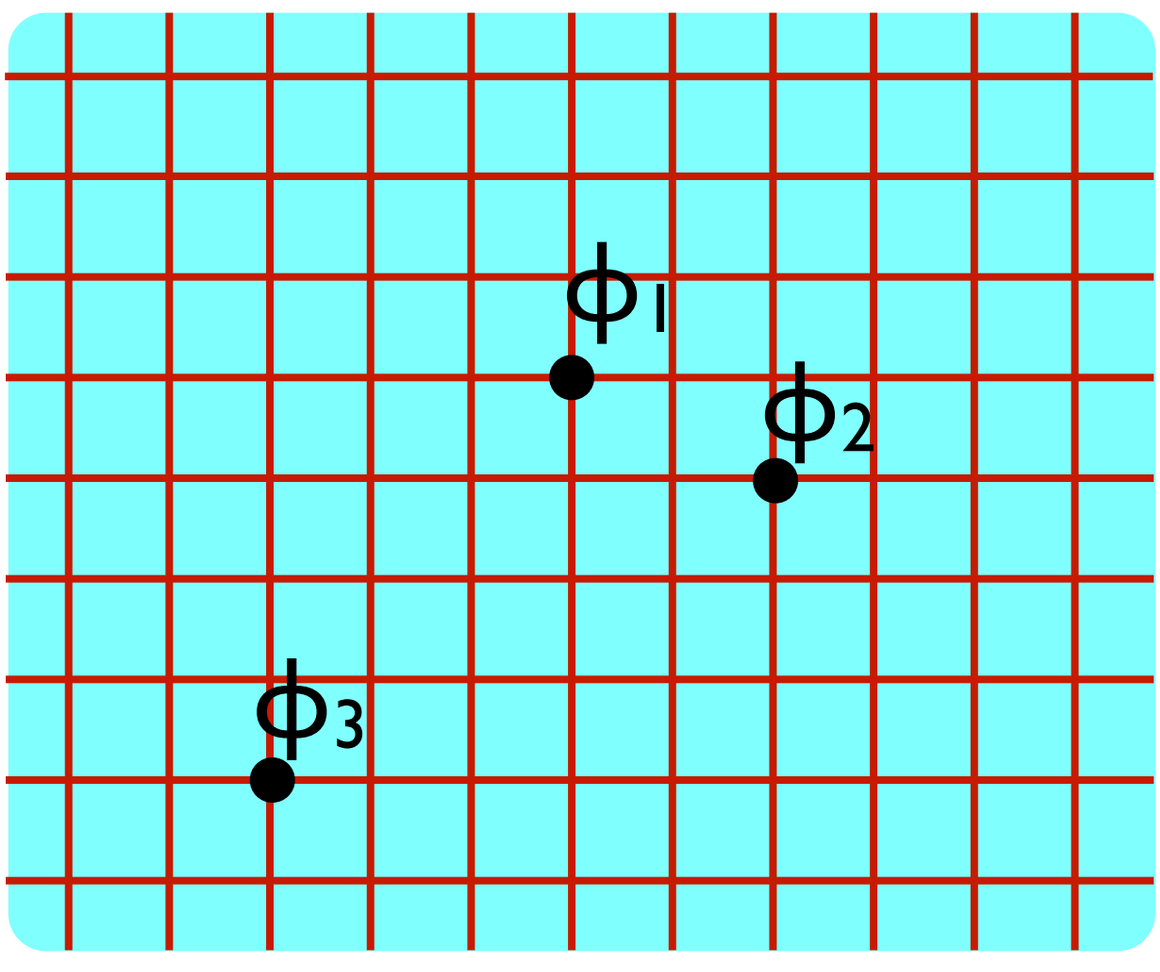
(• The W_{23}-model)

Bulk CFT

Intuitively : continuum limit of lattice model

vector space of lattice fields F (functional on lattice spins)

correlators of lattice fields $\langle \phi_1(z_1)\, \phi_2(z_2)\, \phi_3(z_3) \rangle$

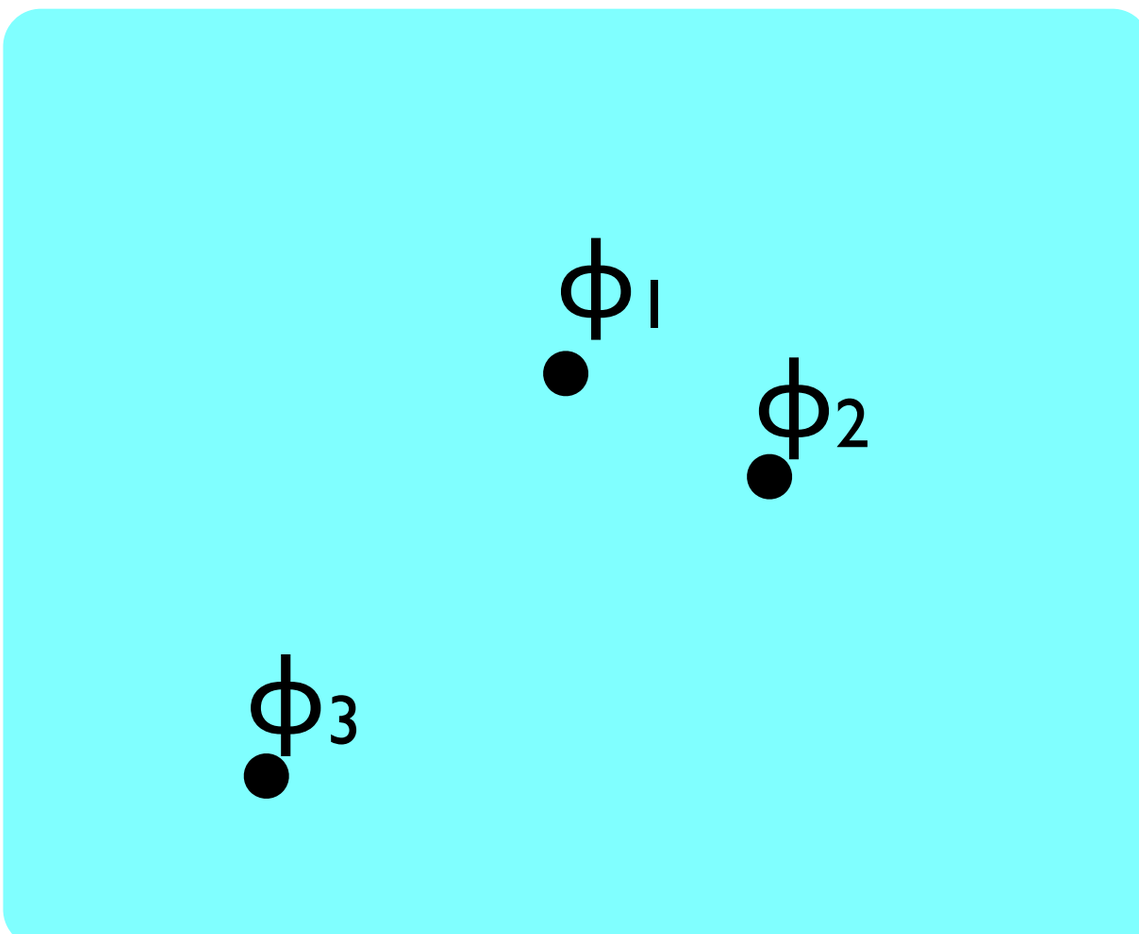


Bulk CFT

Replace by continuum model

vector space of fields F (typically infinite dimensional)

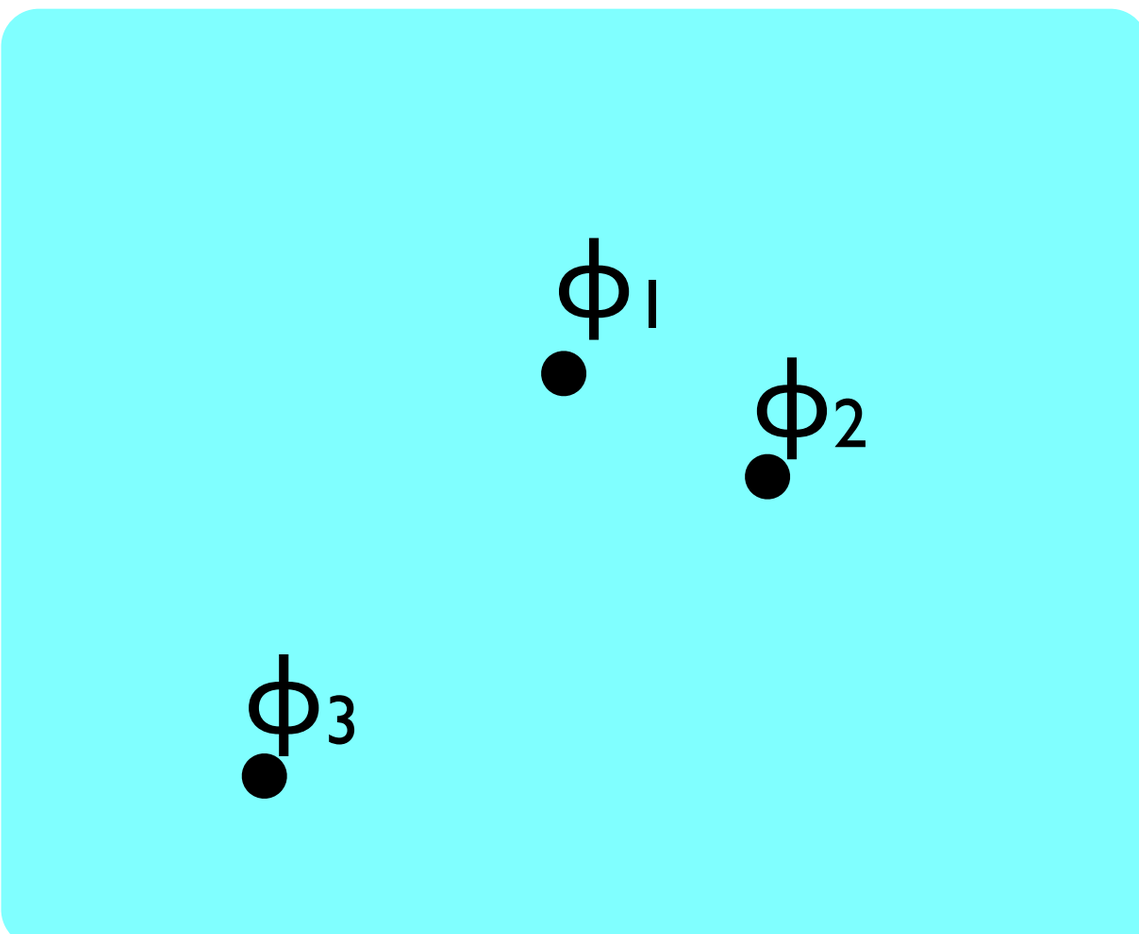
correlators of fields $\langle \phi_1(z_1)\, \phi_2(z_2)\, \phi_3(z_3) \rangle$



. . . bulk CFT

Correlators are functions $(\mathbb{C}^n \setminus \text{diag}) \times F^n \to \mathbb{C}$

Notation : $\langle\, \phi_1(z_1)\, \phi_2(z_2) \ldots \phi_n(z_n) \,\rangle$

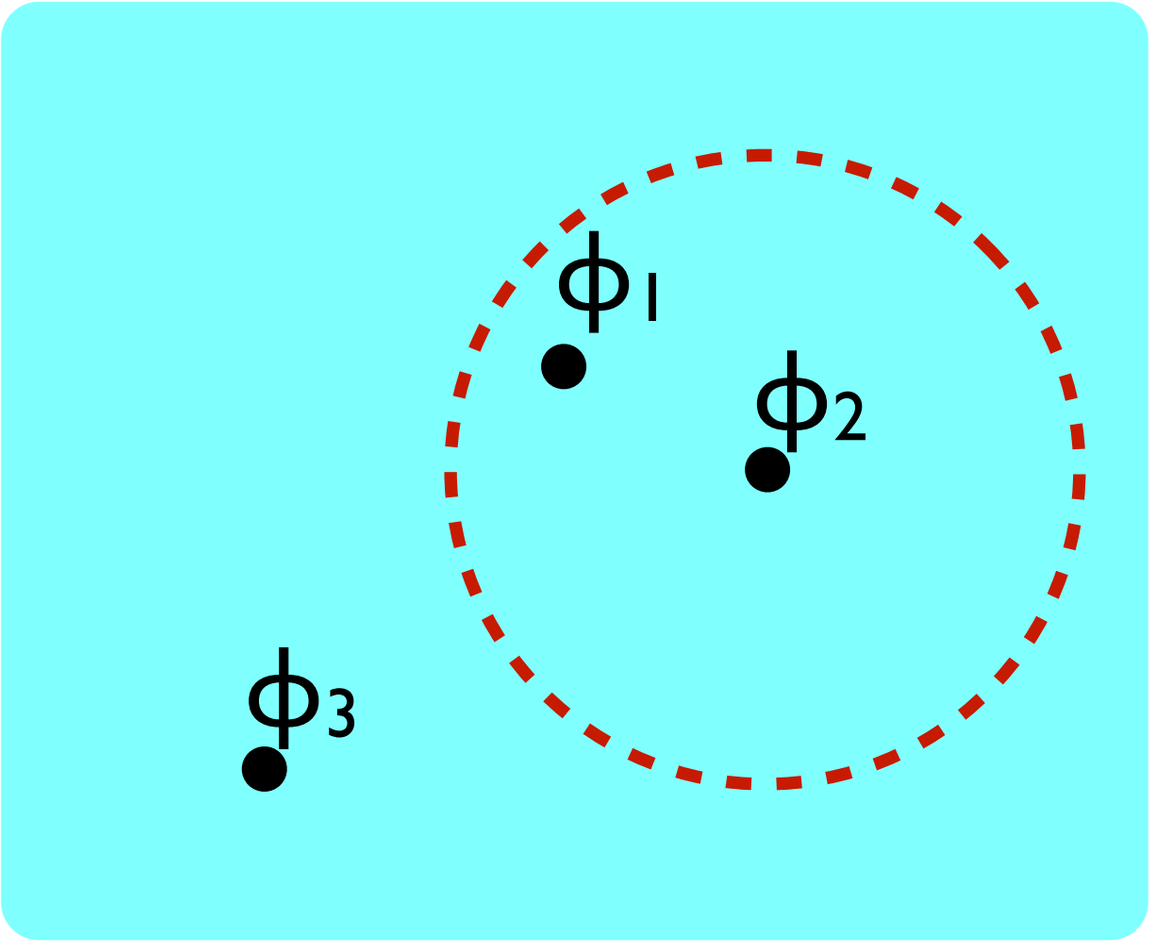


... bulk CFT

Short distance expansion / operator product expansion:

$$\langle \phi_1(z_1)\, \phi_2(z_2)\, \phi_3(z_3) \ldots \phi_n(z_n) \rangle$$

$$= \sum_\alpha f_\alpha(z_1 - z_2) \quad \langle \varphi_\alpha(z_2)\, \phi_3(z_3) \ldots \phi_n(z_n) \rangle$$



This gives a map

$M_x : F \times F \to \overline{F}$,

the bulk OPE

(F is a direct sum of graded components, $\overline{F}$ the direct product)

... bulk CFT

Out-vacuum : $\Omega^* : F \to \mathbb{C}$, $\langle \Omega^*, \phi \rangle = \langle \phi(0) \rangle$

Virasoro action : Virasoro algebra Vir

$sl(2,C) \subset$ Vir , generator L_{-1}, L_0, L_1 , Vir$\oplus$Vir acts on F

Correlators are coinvariants. E.g. 2-pt correlator

$$\langle \phi(x)\, \psi(0) \rangle = \langle \Omega^*, M_x(\phi, \psi) \rangle$$

1) $d/dx \ \langle \Omega^*, M_x(\phi, \psi) \rangle = \langle \Omega^*, M_x(L_{-1}\phi, \psi) \rangle$

2) $\langle \Omega^*, M_x(L_{-1}\phi, \psi) \rangle$

$$= - x^{-1} \langle \Omega^*, M_x \circ (L_0 \otimes id + id \otimes L_0)\, (\phi, \psi) \rangle$$

(+ many more)

…bulk CFT

Logarithms: Solution to differential equation is

$$\langle\, \phi(x)\, \psi(0)\, \rangle$$

$$= \langle \Omega^* \,,\, M_1 \circ e^{-(L_0 \otimes id + id \otimes L_0)\ln x - (\bar{L}_0 \otimes id + id \otimes \bar{L}_0)\ln x^*} (\phi, \psi) \rangle$$

L_0 , $\bar{L}_0$ diagonal : $x^{-h(\phi)-h(\psi)} (x^*)^{-\bar{h}(\phi)-\bar{h}(\psi)}$

L_0 , $\bar{L}_0$ have Jordan blocks : nilpotent part gives ln(x)

Logarithms ⇒ *indecomposable Vir modules* (but not ⇐)

(as exp(2πi L_0) central; in fact: central in any VOA module)

. . . bulk CFT

Data

F , the space of states, a $\mathrm{Vir} \oplus \mathrm{Vir}$-module

$M : \mathbb{C}^\times \times F \times F \to \overline{F}$, the bulk OPE

Ω^* , the out vacuum

Conditions (for theory on complex plane):

- Existence of correlators which are coinvariants and consistent with OPE
- Non-degeneracy of 2-pt correlator

. . . bulk CFT

Non-degeneracy of 2-pt correlator : Let

$$F_0 = \{ \phi \in F \mid \langle \phi(x)\, \psi(0) \rangle = 0 \text{ for all } \psi \in F \}$$

then

- F_0 is independent of x
- F_0 is ideal under OPE
- every correlator vanishes if a single field is from F_0

Can replace F by F / F_0 .

Modular invariance

So far : CFT on complex plane

Can demand : CFT well-defined on a torus.

- 1-point amplitude on torus

$$Z(\phi;\tau) := \mathrm{tr}_F\Big(e^{2\pi i\tau(L_0-c/24)} e^{-2\pi i\tau^*(\bar{L}_0-\bar{c}/24)}\, \phi(0)\Big)$$

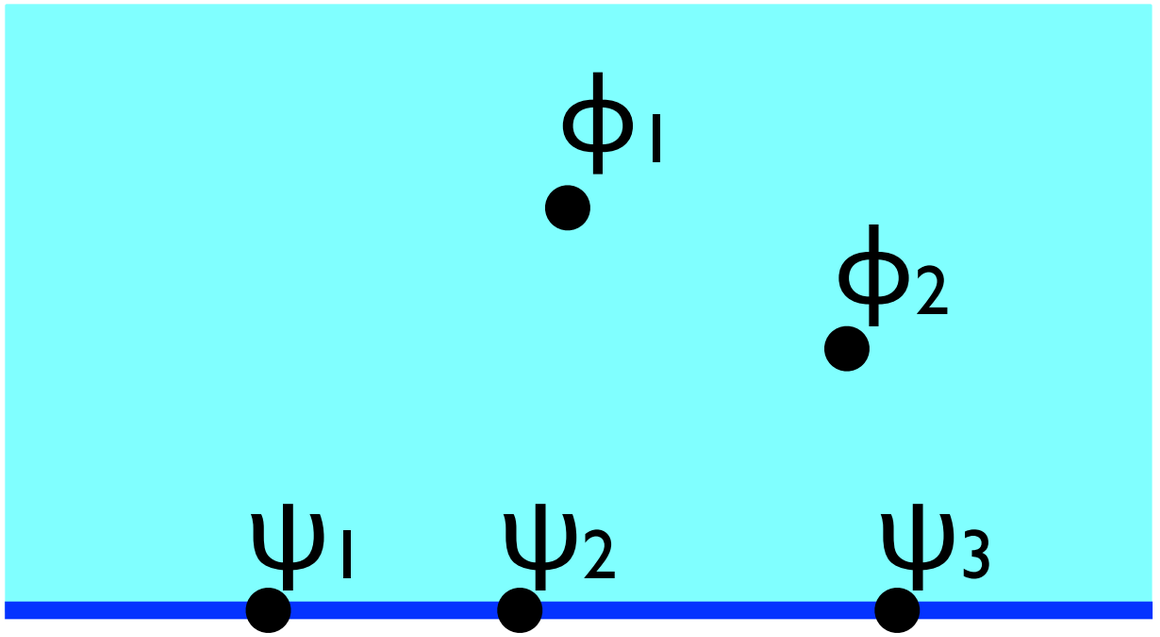
- modular invariance

$$Z(\phi;\tau) = Z(\phi;\tau+1) = Z(\phi';-1/\tau)$$

Rule of thumb :

modular invariant bulk CFTs are all `equally big'.

Boundary CFT



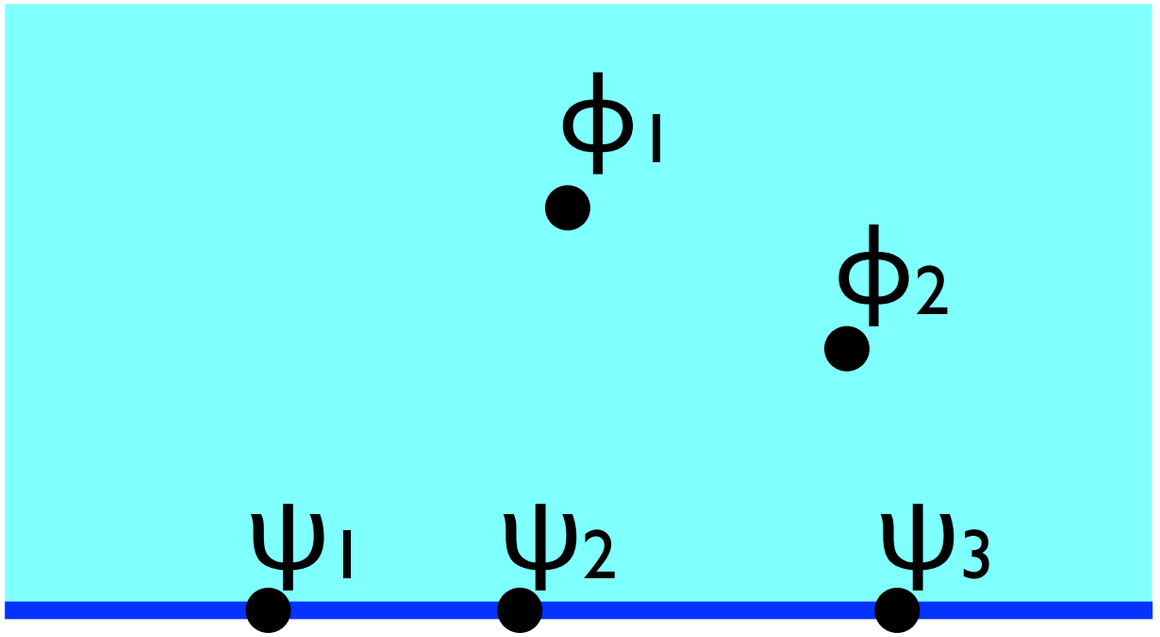
F - space of bulk fields , $\phi_1, \ldots \in F$

B - space of boundary fields, $\psi_1, \ldots \in F$

Correlators on upper half plane

$$\langle \phi_1(z_1)\, \phi_2(z_2) \ldots \psi_1(x_1)\, \psi_2(x_2) \ldots \rangle$$

. . . boundary CFT



Data:

(F, M, Ω^*) , a bulk CFT

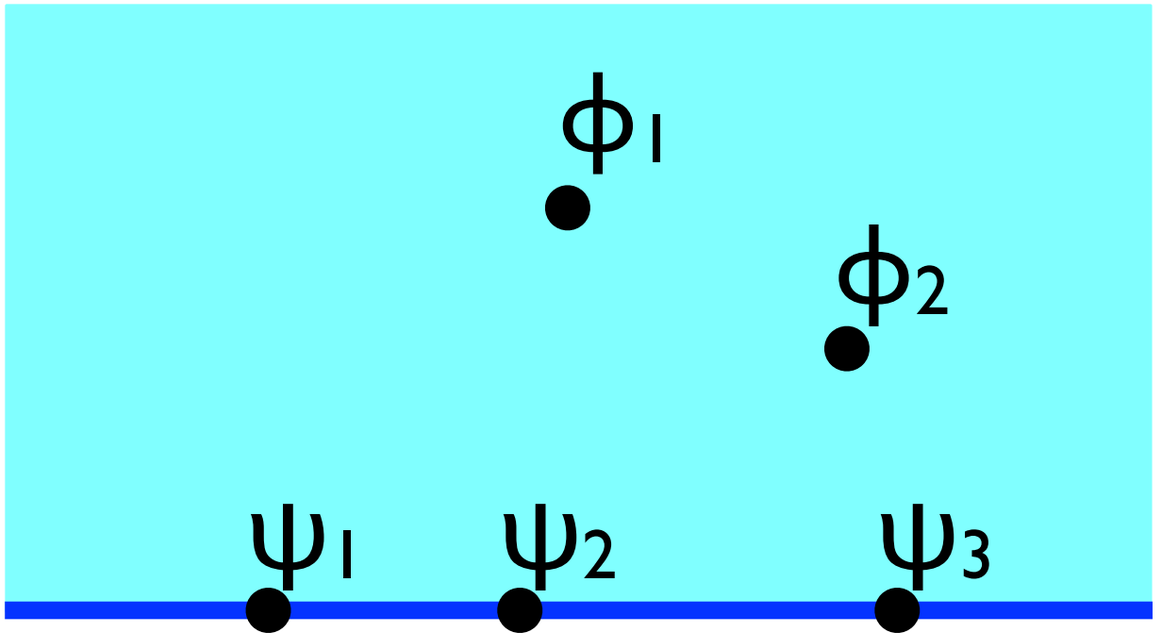
B , boundary fields (a Vir-module)

$\beta_y : F \to \overline{B}$, bulk-boundary OPE

$m_x : B \times B \to \overline{B}$, boundary OPE

$\omega^* : B \to \mathbb{C}$, out-vacuum on upper half plane

. . . boundary CFT



Conditions (for theory on upper half plane):

- Existence of correlators which are coinvariants and consistent with all three OPEs
- Non-degeneracy of 2-pt correlator

… boundary CFT

The basic class of examples : Virasoro minimal models

R_i : finite set of irreducible Vir-modules ($i \in I$)

R_0 : vacuum module - a vertex operator algebra

Cardy case:

$F = \sum_i R_i \otimes \overline{R}_i^*$ (* not necessary in Vir)

$B = R_0$

Other choices of B are $U \otimes_f U^*$

(U is R_0-module, $\otimes_f$ is fusion tensor product)

Note: $R_0 \otimes \overline{R}_0^* \subset F$ is subtheory (closed under OPE), but it is not modular invariant.

From boundary to bulk

Fuchs, Schweigert, IR '02
Gaberdiel, IR '07

Call (B, m, ω^*) a *theory on the boundary.*

Aim: Try to construct (F, M, Ω^*) starting from an 'easy' theory on the boundary.

Idea: Take the `biggest space' F which fits to theory on the boundary.

…from boundary to bulk

Fix a theory on the boundary (B, m, ω^*).

Consider pairs (F', β') such that

0) F' is $\mathrm{Vir}\oplus\mathrm{Vir}$ module (later : $V\otimes V$-module for V a VOA)

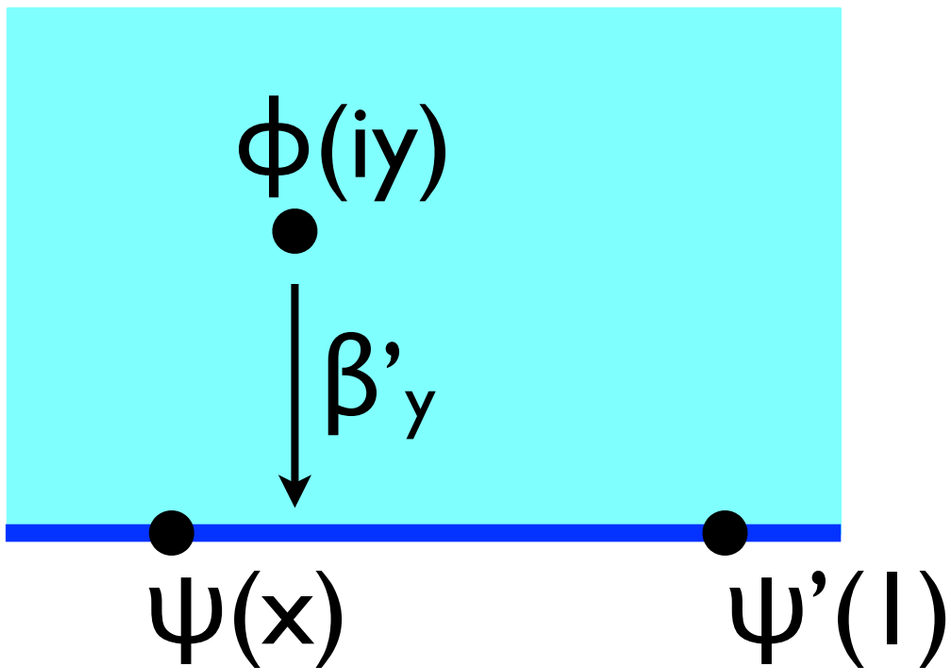
1) $\beta'_y : F \to \overline{B}$ satisfies coinv. condition (for Vir, later for V)

2) β' is central

…from boundary to bulk

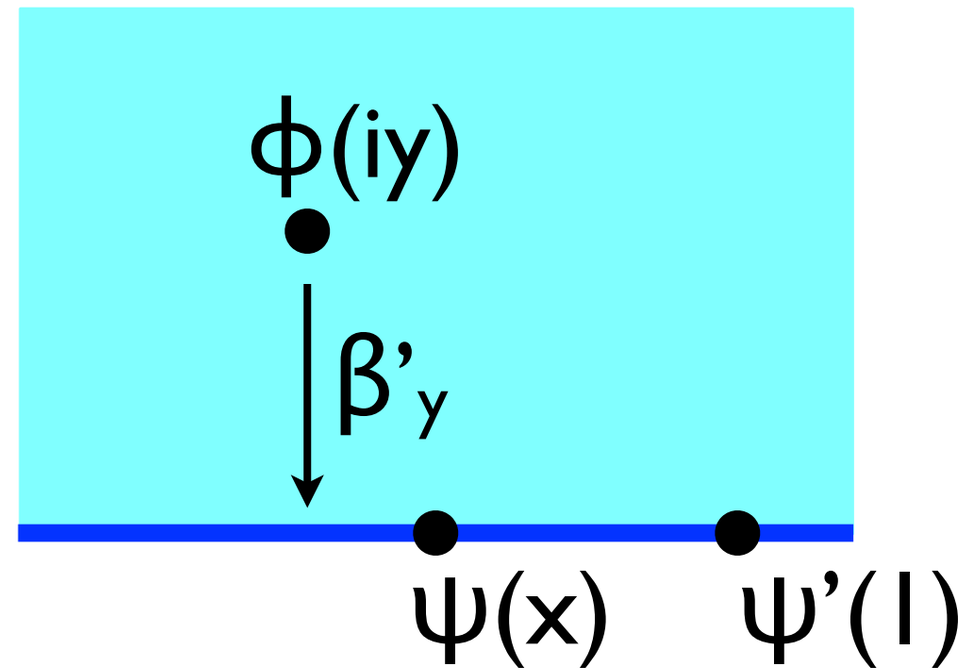
For $x<0$:

$L(x) =$



For $x>0$:

$R(x) =$



β' is *central* if for all $\psi,\psi'\in B$, $\phi\in F'$ and $y>0$:

$$\lim_{x\nearrow 0} L(x) = \lim_{x\searrow 0} R(x)$$

…from boundary to bulk

A morphism of pairs $(F', \beta') \to (F'', \beta'')$ is a $\mathrm{Vir} \oplus \mathrm{Vir}$ intertwiner $f : F' \to F''$ such that the diagram on the right commutes.

$$\begin{tikzcd} F' \arrow[rr, "f"] \arrow[dr, "\beta'_y"'] & & F'' \arrow[dl, "\beta''_y"] \\ & \bar{B} & \end{tikzcd}$$

We make the ansatz that the space of bulk fields $(F(B), \beta(B))$ induced by the theory B on the boundary is the terminal object in the category of such pairs.

If it exists, $(F(B), \beta(B))$ is unique up to unique isomorphism.

…from boundary to bulk

Remark: $\beta(B)_y : F(B) \to \overline{B}$ is injective

Let F_0 be its kernel. By coinv. condition, F_0 is Vir⊕Vir module.

Then the embedding $F_0 \subset F$ is arrow

in the category of pairs.

But also $0 : F_0 \to F$ is an arrow.

By uniqueness, the embedding map

of the kernel is 0.

$$\begin{tikzcd} F_0 \arrow[rr, "\text{embed}"] \arrow[dr, "0"'] & & F(B) \arrow[dl, "\beta(B)_y"] \\ & \bar{B} & \end{tikzcd}$$

…from boundary to bulk

Let (F', β') be another pair with β' injective.

Then $F' \to F(B)$ is injective.

In this sense $F(B)$ is *maximal* space of bulk fields that fits to (B, m, ω^*)

What about M and Ω^* such in data (F, M, Ω^*) for bulk theory? Is the resulting bulk theory modular invariant?

→ algebraic reformulation

Algebra in braided monoidal categories

Let V be a VOA such that Huang-Lepowski-Zhang tensor product theory applies. Then $C = \mathrm{Rep}\, V$ is a $\mathbb{C}$-linear abelian braided monoidal category.

Work with category C (not necessarily $\mathrm{Rep}\, V$) such that (automatic for C=RepV?)

- braided abelian monoidal $\mathbb{C}$-linear
- right exact tensor product
- finite (finite # of isocl. of simple obj., finite dim. Hom)
- $(-)^*$ involutive functor together with a natural iso. $\mathrm{Hom}(A, B) \to \mathrm{Hom}(A \otimes B^*, I^*)$

…algebra in braided monoidal categories

Aside: For $C = \mathrm{Rep}\, V$,

$(-)^*$ is contragredient representation

$\mathrm{Hom}(A, B) \to \mathrm{Hom}(A \otimes B^*, I^*)$ comes from isomorphism of 3pt blocks on P^2 :

- A at 0 , V at x , B^* at ∞
- A at 0 , V at ∞ , B^* at x

Implies existence of internal homs $[A,B] = (A \otimes B^*)^*$.

…algebra in braided monoidal categories

Do *not* assume (all properties below fail in W_{23})

- semi-simple
- tensor unit simple
- rigid
- exact tensor product

Fuchs, Schweigert, IR '02
Kong '06
Gaberdiel, IR '07

... algebra in braided monoidal categories

Translate theory on boundary (B, m, ω^*) to C :

- $B \in C$,
- $m : B \otimes B \to B$ associative product,
- $\varepsilon : B \to I^*$ such that $\varepsilon \circ m$ is a non-degenerate pairing

  (its preimage under $\mathrm{Hom}(B, B^*) \to \mathrm{Hom}(B \otimes B, I^*)$ is an isomorphism)
- demand in addition choice of unit $\eta : I \to B$

  ($C = \mathrm{Rep}\, V$: boundary condition respects V-symmetry)

…algebra in braided monoidal categories

Translate pairs (F', β') to C :

Notation:

- $\bar{C}$ is C with inverse braiding
- $C \boxtimes \bar{C}$ is Deligne product
- $T : C \boxtimes \bar{C} \to C$ is induced by tensor product
  (takes $U \boxtimes V$ to $U \otimes V$ - uses right exactness of $\otimes$)

Then $F' \in C \boxtimes \bar{C}$, $\beta' : T(F') \to B$.

Kong '06
Gaberdiel, IR '07

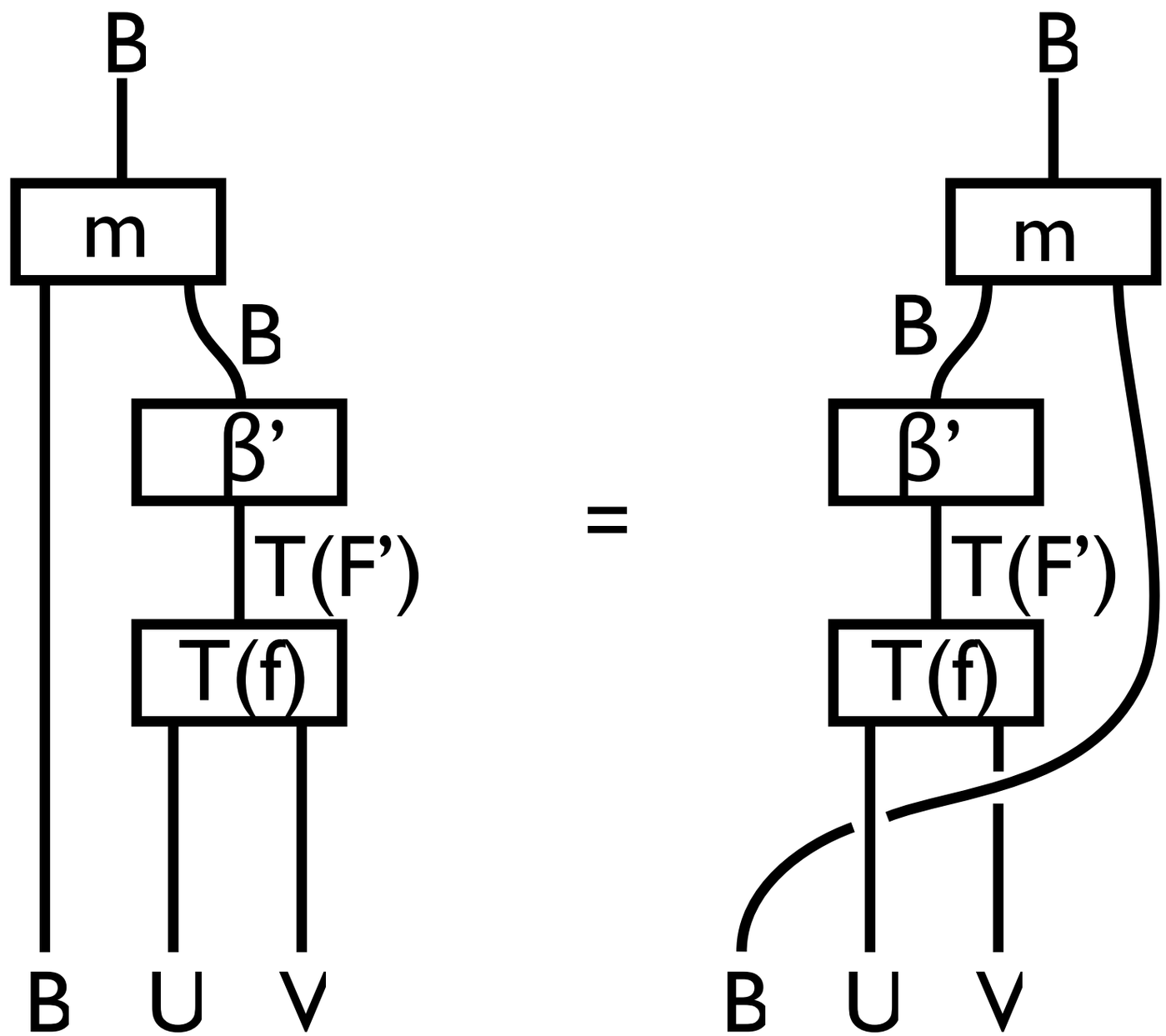
. . . algebra in braided monoidal categories

Translate centrality condition for $\beta' : T(F') \to B$:

For all

$f : U \boxtimes V \to F'$

have



Fjelstad, Fuchs, Schweigert, IR '06
Davydov '09

...algebra in braided monoidal categories

Def: Let M be a monoidal category and let (A,m,η) be an algebra in M. The *full centre* Z(A) is an object in the monoidal centre $\mathcal{Z}$(M) which is terminal among pairs (Y∈$\mathcal{Z}$(M) , u:Y→A) such that

$$\begin{tikzcd}
Y \otimes A \arrow[r, "u\otimes id"] \arrow[d, "y_A"] & A \otimes A \arrow[dr, "m"] & \\
A \otimes Y \arrow[r, "id\otimes u"] & A \otimes A \arrow[r, "m"] & A
\end{tikzcd}$$

commutes.

Note: Z(A) may not exist.

…algebra in braided monoidal categories

Thm: Davydov '09

Suppose $Z(A)$ exists.

Then there exists a unique unital algebra structure $(Z(A), m_z, \eta_z)$ on $Z(A)$ such that $u : Z(A) \to A$ is a unital algebra map.

With this algebra structure, $Z(A)$ is commutative.

…algebra in braided monoidal categories

There is a braided monoidal functor $E : C \boxtimes \overline{C} \to \mathcal{Z}(C)$.

If E is an equivalence, the terminal object $(F(A), \beta(A))$, with $F(A) \in C \boxtimes \overline{C}$, is isomorphic to $Z(A)$.

In any case, there is a unique commutative algebra structure on $F(A) \in C \boxtimes C$ such that $\beta(A) : T(F(A)) \to A$ is an algebra map.

Non-logarithmic rational CFT

Huang '05

$\mathcal{C} = \mathrm{Rep}\, V$ is a modular category.

Fuchs, Schweigert, IR '02
Fjelstad, Fuchs, Schweigert, IR '06
Kong, IR '08

Theory on the boundary is a symmetric Frobenius algebra A in $\mathcal{C}$.

The bulk theory is the full centre $Z(A)$, a commutative symmetric Frobenius algebra in $\mathcal{C} \boxtimes \overline{\mathcal{C}}$.

$Z(A)$ is modular invariant.

E.g. : $Z(I) = \sum_i U_i \boxtimes U_i^*$

… non-logarithmic rational CFT

Thm:

All modular invariant commutative Frobenius algebras Z with $\dim \mathrm{Hom}(1,Z)=1$ are given by $Z(A)$ for some Frobenius algebra A.

Furthermore, $Z(A) = Z(B)$ iff A and B are Morita equivalent.

W_{1p} models

Kausch '91, Gaberdiel, Kausch '96
Fuchs, Hwang, Semikhatov, Tipunin '03
Adamovic, Milas '07
...

$C = \mathrm{Rep}\, V$ is (conjecturally?) a finite tensor category.

Quella, Schomerus '07
Gaberdiel, IR '07

- $F(1) = \sum_i P_i \boxtimes P_i^* / N$

- as $\mathbb{R} \times \mathbb{R}$ graded vector space have $F(1) = \sum_i U_i \boxtimes P_i^*$
(graded by generalised L_0 and $\overline{L}_0$ eigenspaces)

- $F(1)$ gives a modular invariant torus partition function.

The W_{23} model

Feigin, Gainutdinov, Semikhatov, Tipunin '06

- Virasoro Verma module for h=0 and c=0: two independent null vectors

$$N_1 = L_{-1}\Omega \qquad\qquad N_2 = (L_{-2} - \tfrac{3}{2}L_{-1}L_{-1})\Omega$$

- Divide by N_1 and N_2 : get

$$\mathcal{V}(0) = \mathbb{C}\cdot\Omega$$

- Divide by N_1 but <u>not</u> by N_2 : get V with character

$$\chi_{\mathcal{V}}(q) = 1 + q^2 + q^3 + 2q^4 + 2q^5 + 4q^6 + 4q^7 + 7q^8 + 8q^9 + 12q^{10} + 14q^{11} + 21q^{12} + 24q^{13} + 34q^{14} + 41q^{15} + 55q^{16} + \cdots$$

(quasi-rational, but not rational)

. . . the W_{23} model

- Extend by 3 fields with h=15, get W with character

$$\chi_{\mathcal{W}}(q) = 1 + q^2 + q^3 + 2q^4 + 2q^5 + 4q^6 + 4q^7 + 7q^8 + 8q^9 + 12q^{10} + 14q^{11} + 21q^{12} + 24q^{13} + 34q^{14} + 44q^{15} + 58q^{16} + \cdots$$

$$\left(\begin{array}{l} \chi_{\mathcal{V}}(q) = 1 + q^2 + q^3 + 2q^4 + 2q^5 + 4q^6 + 4q^7 + 7q^8 + 8q^9 + 12q^{10} \\ \qquad + 14q^{11} + 21q^{12} + 24q^{13} + 34q^{14} + 41q^{15} + 55q^{16} + \cdots \end{array} \right)$$

- W is indecomposable but not irreducible

$$0 \longrightarrow \mathcal{W}(2) \longrightarrow \mathcal{W} \longrightarrow \mathcal{W}(0) \longrightarrow 0$$

irreducible sub-representation

irreducible quotient

(Does not happen for the $W_{l,p}$ models.)

. . . the W_{23} model

$C = \mathrm{Rep}\, V$ supposedly

- abelian, braided, monoidal, $\otimes$ right exact, $(-)^*$

but <u>not</u>

- semi-simple, rigid, simple I, $\otimes$ exact

. . . the W_{23} model

Feigin, Gainutdinov, Semikhatov, Tipunin '06
Adamovic, Milas '09

Irreducible representations are:

	$s=1$	$s=2$	$s=3$
$r=1$	$0,\ 2,\ 7$	$0,\ 1,\ 5$	$\frac{1}{3},\ \frac{10}{3}$
$r=2$	$\frac{5}{8},\ \frac{33}{8}$	$\frac{1}{8},\ \frac{21}{8}$	$-\frac{1}{24},\ \frac{35}{24}$

Gaberdiel, Wood, IR '09

The tensor unit W does not have a non-degenerate pairing. The simplest theory on the boundary is:

$$B := W(5/8) \otimes_f W(5/8) = \begin{tikzcd} & 2 \arrow[dl] \arrow[d] \arrow[dr] & \\ 7 \arrow[dr] & 0 \arrow[d] & 7 \arrow[dl] \\ & 2 & \end{tikzcd}$$

. . . the W_{23} model

Gaberdiel, Wood, IR '10

$$B := W(5/8) \otimes_f W(5/8) = \begin{tikzcd} & 2 \arrow[dl] \arrow[d] \arrow[dr] & \\ 7 \arrow[dr] & 0 \arrow[d] & 7 \arrow[dl] \\ & 2 & \end{tikzcd}$$

We could not compute $F(B)$ for this algebra.

Instead, we computed (but without full proof) $F(W^*)$ of the commutative algebra W^*.

Main ingredient : conjectural composition series of conjectural projective cover $P(0)$ of $W(0)$, using results of Adamovic, Milas '10

. . . the W_{23} model

Composition series of $F(W^*)$ in integer weight-sector

	0	1	2	5	7
0	1				
1		1			
2			1		
5				1	
7					1

$\longrightarrow$

	0	1	2	5	7
0		1	1		
1	1			2	2
2	1			2	2
5		2	2		
7		2	2		

$\longrightarrow$

	0	1	2	5	7
0	1			2	2
1		2	4		
2		4	2		
5	2			2	4
7	2			4	2

$\longrightarrow$

	0	1	2	5	7
0		1	1		
1	1			2	2
2	1			2	2
5		2	2		
7		2	2		

$\longrightarrow$

	0	1	2	5	7
0	1				
1		1			
2			1		
5				1	
7					1

.

$F(W^*)$ has a modular invariant partition function and is a commutative algebra. But it has no unit (just as W^*).

谢谢大家！